Locally-responsible Artificial Intelligence frameworks: Designing a Digital/AI Toolkit Empowering Community-led Digital Data Governance of Cultural Heritage in Burkina Faso

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Abstract:
As the digital era evolves with the emergence of widely-available generative artificial intelligence (AI) technologies, safeguarding and promoting African intangible cultural heritage (ICH) presents both opportunities and challenges. While digital technologies and AI offer tools for promotion and innovation, they also pose risks of misrepresentation, exploitation, and cultural erasure—risks amplified by the global underrepresentation of African datasets and heritage online. Rights-based and human-centric digital data governance could contribute to sustainable cultural and economic development for artists and local communities (Effoduh et al. 2024; Shelver et al. 2024).

African cultural content, particularly from traditional artist communities, is currently underrepresented online, which can be a source of ignorance and misunderstanding (Hassan 2022). Redressing the North-South data imbalance is as important as providing equitable internet access, and access to AI tools, because there may be a significant cost associated with online invisibility in the long term. However, online underrepresentation may also give African artists and cultural creators a temporary competitive advantage over western contemporary and traditional artists at the moment, since their work currently cannot be as easily copied and used by third parties using generative AI. Nevertheless, African creators will likely seek to both protect their cultural heritage and business interests, while also engaging in cultural promotion online and using digital tools for innovation (Shelver et al. 2024).

This short paper presents the framework of a critical digital literacy toolkit being designed and developed as part of a British Academy-funded research project involving a research team from the UK and Burkina Faso. The project aims to support community digital data governance strategies that promote response-ability (Klumbyte et al. 2023): the ability to respond to new digital technologies such as AI when sharing cultural data online. The toolkit aims to assist ICH practitioners and communities who have not yet shared significant amounts of cultural data online to reflect on future data sharing practice by creating a space for discussing and addressing digital literacy needs.

The toolkit seeks to address a gap in basic digital/AI skills training in the creative sector and elsewhere, which is often focused on individuals rather than communities, lacking in discussion of the ethics of collective data governance, and oriented towards western audiences with good digital access. It draws on dialogues about language representivity and community data licensing for African languages (for example the NOODL Nwulite Obodo Open Data License), and existing reports and critical AI literacy resources aimed at African artists in the cultural industries sector developed by organizations such as Creatives Garage in Kenya and Electric South in South Africa. However, there are currently few capacity-building resources on digital cultural heritage data governance for African communities at the beginning of their digital journey.

The proposed toolkit framework has been informed by initial discussions with community lea-

ders of the basket weavers and NGOs for mask society members and musicians in Burkina Faso. It also draws on Dr Leonce Ki's prior work with mask society members in Burkina Faso. Community members have expressed interest in controlling the sharing of knowledge about mask society practices, managing third-party photography at mask festivals and using digital tools for innovation, for example in basket design. The content of the toolkit has been designed to address some of these needs and will be further developed, refined and tailored in close collaboration with ICH practitioners from these communities after some logistical delays to the in-depth work in Burkina Faso have been addressed. A group of African experts will also be invited to workshop the toolkit content, in developing a version adapted for broader audiences such as policymakers and culture sector stakeholders.

The toolkit framework incorporates four main themes - Awareness, Promotion, Innovation and Protection - to address diverse aspects of community data governance for ICH in the digital age. The awareness theme of the toolkit aims to cover a basic introduction to digital technologies and AI use in relation to ICH. It will explain how AI systems are trained using large datasets of cultural data and thereby generate new content. It will also provide information about the opportunities and risks of sharing cultural data online in this context, highlighting the impact on cultural practitioners and communities.

The innovation theme will focus on ways in which cultural practitioners can incorporate AI tools into their creative processes while respecting cultural traditions and norms. For example, translation tools could be used to communicate with customers, and musicians could for example use AI to analyse traditional rhythms and generate modern compositions that blend heritage with contemporary trends. An art project by artist Linda Dounia from Senegal based on flowers from the region will be referenced (with permission) to illustrate the possibilities and limitations of AI model training (Dounia 2021). Existing AI tools are often biased towards western sources and must be paired with strategies to prevent misuse, such as unauthorised reproduction or uncredited distribution, which remain pervasive issues in the digital space.

The promotion and protection themes will guide cultural heritage practitioners in promoting their work and cultural practices online while exerting some control over how it is used, attributed and accessed by others. The promotion theme will give examples of heritage-sensitive digital marketing practices to promote cultural products and services, and explain some strategies for social media platform selection, storytelling, and use of hashtags. Attribution and ethical sharing of cultural data can be encouraged through creative commons licenses, local context labels and ethical guidelines for tourists and other third parties. The protection theme will focus on ways in which communities and practitioners effectively shape their own data governance strategies to decide what kinds of data to share online, and protect data shared online against unauthorised data scraping, for example opt-outs, glazing and watermarks.

Since many of the ICH practitioners and communities working with the project do not yet have stable internet connections, the project will develop a low-tech offline version of the toolkit as well as one that requires a live internet connection. Given that people in Burkina Faso speak a number of local languages, and French in some cases, the toolkit will focus on low-text explanations, and interactive engagement to maximise reach and benefit. An English version of the toolkit will be shared online as well as part of the project.

The project emphasises responsible AI usage and community data governance, following the CARE principles (Collective Benefit, Authority to Control, Responsibility, and Ethics, Carroll et al. 2020), for example, in toolkit design and development. This approach is informed by the experiences of Indigenous communities in various countries and regions (e.g. Fitch et al. 2023), although the communities working with the project in Burkina Faso do not self-identify as Indigenous.

No cultural data will be shared without the free, prior and informed consent of the practitioners and their communities, based on a community data governance protocol. The project will not use community data to train any AI models, maintaining community control and ownership over their data and illustrating the toolkit where appropriate with selected quotes and a limited number of community pictures used with free, prior and informed consent and attribution shared under a CC BY NC ND license.